

MTH208: Problem Set

Problem Set Questions

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Practice R Questions

1. Define an R vector of the first 1000 odd integers.
2. Define an R vector of the first 500 Fibonacci numbers.
3. Write an R function that rolls a fair die and returns 1 if the die turns out to be even.
4. Write an R function that tosses a fair coin 15 times. The function should return "lose" if the number of heads is less than 8 and "win" if the number of heads is more than or equal to 8.
5. Define a 5×5 matrix of all elements being 1.
6. Define a 5×5 diagonal matrix with the diagonals being 1:5.
7. Define a 10×10 matrix where each entry is the result of a random roll of a fair die.
8. Write an R function that takes inputs `n` and `rho`. The function should return an $n \times n$ matrix with 1s on the diagonals and `rho` on the off-diagonals.
9. Write an R function that takes inputs `n` and `rho`. The function should return an $n \times n$ matrix with the i, j the element being $\rho^{|i-j|}$.
10. Write an R function that takes a matrix input and returns a smaller matrix with only the odd columns of the original matrix.
11. Define an 4 dimensional arrays with dimensions $10 \times 4 \times 6 \times 5$ where the entries are all 1s.

Worksheet Based Questions

Basic and Simulation Questions

1. Write an R function to calculate the area of a circle of radius r for a user-given value of r .
2. Write an R function that returns the larger of two inputs x and y .

- Write an R program that saves the output of 1000 rolls of a fair die and returns the number of time the output was an even number.
- Draw 1000 random number between $[0, 1]$ and calculate the proportion of numbers between 0.1 and 0.2.
- A popular chips company came out with a new strategy. They were going to put different Harry Potter toys in their chips packets. You and your friends are very excited; you want to be the first to collect all figurines! The chips company has declared that they will put the toys in the packets based on the popularity of the character. So more popular characters are more likely to be in the packet. Below in the table is the probability of a packet of chips containing the type of toys.

Toys	Harry	Dumbledore	Hermione	Ron	Neville	Mcgonagall	Dobby
Prob	.25	.20	.20	.15	.10	.05	.05

Write an R function to calculate the number of chips packets that need to be bought to collect all toys. Repeat the simulation 1000 times and calculate the average number of chips packets that need to be bought to collect all toys.

- You take half of a vitamin every morning. The vitamins are sold in a bottle of 100 (whole) tablets, so at first you have to cut the tablets in half. Every day you randomly pull one thing from the bottle. If it's a whole tablet, you cut it in half and put the leftover half back in the bottle. If it's a half-tablet, you take the vitamin. You just bought a fresh bottle. How many days, on average, will it be before you pull a half-tablet out of the bottle?
- You are on a game show, being asked to choose between three doors. One door has a car, and the other two have goats. After you choose a door, the host, Monty Hall, opens one of the other doors, which he knows has a goat behind it. Monty then asks whether you would like to switch your choice of door to the other remaining door. Do you choose to switch or not to switch?

Whether you switch or not depends on which action has the largest probability of winning the car.

- Write an R function, `MontyHall()` to simulate this game show for when the contestant decides to switch doors. The R function should return 1 if the contestant wins and 0 if they lose.
- Repeat the simulation 1000 times to estimate the probability of winning if the contestant switches.
- Write an R function to simulate a random walk of n steps. Each step is determined by a fair coin toss: heads means step forward (+1), tails means step backward (-1). The function should return the final position after n steps. Repeat the simulation 1000 times for $n=100$ steps and calculate the average final position (beware of the first and last step, you cannot move before first and after last step).

9. Write an R function to simulate the following scenario: In a large city, people visit a popular mall on random days throughout the year. The function should estimate the probability that at least two people in a group of `n` people visit the mall on the same day. Assume there are 365 days in a year. Repeat the simulation 1000 times for `n=23` and calculate the estimated probability.
10. Write an R function that simulates a geometric random variable with a given probability of success `p`. The function should return the number of trials needed to get the first success. Repeat the simulation 1000 times and calculate the average number of trials needed.

Image Questions

1. Write a function `prop.color` that calculates the proportion of pixels in a given image that are within a 0.5 Euclidean distance from a given color. That is, if x_{ij} is the rgb vector of the i, j th pixel and c is a given color vector, then function should return the proportion of pixels in the image for which

$$\|x_{ij} - c\| \leq 0.5$$

(Here norm is the 2-norm or the Euclidean norm).

The function should take two arguments; `img` which will be an `imager` image and `col` which will take a vector of length 3.

2. Write R code that takes an image and produces a mirrored image.
3. Write R code that replaces all color vectors less than `(.10, .10, .10)` with `(0, 0, 0)`.
4. Construct a matrix `A` which is 1000x1000. Find the norm of every column of the matrix using a loop and then also using function `sapply`. Which is faster?
5. Write a code to make a water water reflection of a given image and plot the images one below the other.
 - a. Think of which axis will the reflection work
 - b. Can you add some things to the reflected image to make it seem like a water reflection? (This is not easy, but try)
6. Find the mirror reflection of given image and plot the images side by side.
7. Cut an image in two vertical halves, and swap them.
8. Cut an image into to horizontal halves and swap them.
9. In the given imager `img` , write R function `find.replace()` that find out the least pure color `fcol` and replace it by given `rcol`. Do this by :
 - a. Without using for loops
 - b. With for loops

Which one will be faster?

10. Write a R function `expand(img,r)` to expand the given image to $r \times r$ dimension image, where image is smaller than $r \times r$. There are many ways to do this. Try whatever makes sense to you

Webscraping

1. Scrape the link: “<https://www.espn.com/tenis/rankings/_/tipo/wta>” and make a clean table.
2. Extract the names, years of birth, and years of death of the top 10 scientists from the website [Top-10 Scientists](#). Determine the number of scientists who were alive at any point between 1830 and 1880, create a histogram showing the number of scientists alive in each year, and identify the year with the highest number of living scientists.
3. Scrape the complete table of summer olympic medals of the United States available at https://en.wikipedia.org/wiki/United_States_at_the_Olympics. It may be challenging to get everything, but try as much as possible.
4. From the website:
<https://stats.stackexchange.com/questions?tab=Votes>, scrape the following information:
 1. The title of the questions
 2. The number of views
 3. The number of answers
 4. The notes of votes